

2-RINGS

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ABSTRACT. These are the notes for the third talk in the seminar *Higher Zariski Geometry*.

1. PRESENTABLE (STABLE) CATEGORIES

Remark 1.1. Recall that a cardinal κ is called *regular* if the category of sets of cardinality $< \kappa$ has all colimits of size $< \kappa$. All the cardinals appearing in these notes are assumed to be regular. Recall that a category is called κ -*small* if its underlying space is κ -small and every homotopy group of its mapping spaces are κ -small. Recall also that a category I is κ -*filtered* if every diagram $K \rightarrow I$ from a κ -small category can be extended to a cocone diagram $K^{\triangleright} \rightarrow I$.

Construction 1.2. Let $\mathcal{C} \in \mathbf{Cat}$ be a category. Let $\mathbf{PSh}(\mathcal{C}) = \mathbf{Fun}(\mathcal{C}^{\mathrm{op}}, \mathbf{Spc})$ be the presheaf category and let $\mathfrak{y}_{\mathcal{C}} : \mathcal{C} \rightarrow \mathbf{PSh}(\mathcal{C})$ be the Yoneda embedding. The Ind_{κ} -*completion* of $\mathcal{C}$ is defined as the full subcategory

$$\mathrm{Ind}_{\kappa}(\mathcal{C}) \subseteq \mathbf{PSh}(\mathcal{C}),$$

on those presheaves which are κ -filtered colimits of representables. Notice that the Yoneda embedding factors through a fully-faithful functor $\mathfrak{y} : \mathcal{C} \hookrightarrow \mathrm{Ind}_{\kappa}(\mathcal{C})$. One can show, by restricting the universal property of the presheaf category, that the Ind_{κ} -completion is such that restriction along the Yoneda embedding induces an equivalence

$$\mathfrak{y}^* : \mathbf{Fun}^{\kappa\text{-filtered}}(\mathrm{Ind}_{\kappa}(\mathcal{C}), \mathcal{D}) \rightarrow \mathbf{Fun}(\mathcal{C}, \mathcal{D}),$$

for every category $\mathcal{D}$ with κ -filtered colimits. Here the left hand side denotes the full subcategory of $\mathbf{Fun}(\mathrm{Ind}_{\kappa}(\mathcal{C}), \mathcal{D})$ spanned by those functors preserving κ -filtered colimits.

Remark 1.3. Let $\mathcal{C} \in \mathbf{Cat}^{\kappa\text{-filtered}}$ be a category with κ -filtered colimits. Recall that an object $x \in \mathcal{C}$ is called κ -*compact* if mapping out of it preserve κ -filtered colimits. The collection of κ -compact objects organize into a full subcategory $\mathcal{C}^{\kappa} \subseteq \mathcal{C}$ which is in general closed under κ -small colimit, as κ -small limits commute with κ -filtered colimits in $\mathbf{Spc}$. If $\mathcal{C} = \mathrm{Ind}_{\kappa}(\mathcal{D})$ for a small category $\mathcal{D} \in \mathbf{Cat}$, then there is an inclusion $\mathcal{D} \subseteq (\mathrm{Ind}_{\kappa}(\mathcal{D}))^{\kappa}$ which is not an equality in general.

Definition 1.4. We say that a category is κ -*presentable* if it is of the form $\mathrm{Ind}_{\kappa}(\mathcal{C})$ for $\mathcal{C} \in \mathbf{Cat}^{\mathrm{rex}(\kappa)}$ a small category with κ -small colimits. We let $\mathbf{Pr}_{\kappa}^{\mathrm{L}} \subseteq \widehat{\mathbf{Cat}}$ be the non full subcategory of κ -presentable categories and colimit preserving functors that preserve the κ -compact object.

Remark 1.5. Let $\lambda \geq \kappa$ be regular cardinal. Then every λ -filtered category is κ -filtered, so that every κ -compact object is a λ -compact object. Furthermore, every κ -presentable category is λ -presentable. Indeed, if $\mathcal{C} \simeq \mathrm{Ind}_{\kappa}(\mathcal{C}^{\kappa})$ with $\mathcal{C}^{\kappa}$ small and admitting κ -small colimits, then every object of $\mathcal{C}^{\kappa}$ is κ -compact, hence λ -compact. Thus $\mathcal{C}^{\kappa} \subseteq \mathcal{C}^{\lambda}$. Since $\mathcal{C}^{\kappa}$ generates $\mathcal{C}$ under κ -filtered colimits, and every κ -filtered colimit may be refined by λ -filtered colimits after enlarging the compact subcategory to $\mathcal{C}^{\lambda}$, the canonical functor

$$\mathrm{Ind}_{\lambda}(\mathcal{C}^{\lambda}) \rightarrow \mathcal{C}$$

is an equivalence, making $\mathcal{C}$ is λ -presentable.

Remark 1.6. One can work with all κ -presentable categories at the same time by defining the category of *presentable categories* as the colimit

$$\mathrm{Pr}^L = \mathrm{colim}_{\kappa} \mathrm{Pr}_{\kappa}^L$$

along the inclusion $\mathrm{Pr}_{\kappa}^L \hookrightarrow \mathrm{Pr}_{\lambda}^L$ for $\lambda \geq \kappa$.

Arrows of presentable categories are well behaved:

Remark 1.7. Let $f : \mathcal{C} \rightarrow \mathcal{D}$ be a functor between categories in Pr^L . Then f is a left adjoint if and only if it preserves small colimits, and it is a right adjoint if and only if it preserves small limits and is accessible, that is it preserves κ -filtered colimits for some regular cardinal κ .

We now turn to the commutative algebra of Pr^L .

Remark 1.8. Lurie constructed in [Lur17, Section 4.8] a symmetric monoidal structure on Pr^L characterized by the following universal property: given two presentable categories $\mathcal{C}$ and $\mathcal{D}$, their Lurie tensor product is a presentable category $\mathcal{C} \otimes \mathcal{D}$ with a functor $\mathcal{C} \times \mathcal{D} \rightarrow \mathcal{C} \otimes \mathcal{D}$ such that for every presentable category $\mathcal{E}$, precomposition with it induces an equivalence

$$\mathrm{Fun}^L(\mathcal{C} \otimes \mathcal{D}, \mathcal{E}) \rightarrow \mathrm{Fun}^{L,L}(\mathcal{C} \times \mathcal{D}, \mathcal{E}).$$

Here the superscript L, L denotes the full subcategory spanned by those functors $\mathcal{C} \times \mathcal{D} \rightarrow \mathcal{E}$ which preserve colimits separately in each variable. It follows from this universal property that the symmetric monoidal structure is also closed. To be precise, for every pair of presentable categories $\mathcal{C}$ and $\mathcal{D}$, there exists a natural equivalence

$$\mathrm{Fun}^L(\mathcal{C} \otimes \mathcal{D}, \mathcal{E}) \simeq \mathrm{Fun}^{L,L}(\mathcal{C} \times \mathcal{D}, \mathcal{E}) \simeq \mathrm{Fun}^L(\mathcal{C}, \mathrm{Fun}^L(\mathcal{D}, \mathcal{E}))$$

in $\mathcal{E} \in \mathrm{Pr}^L$, so that $\mathrm{Fun}^L(-, -)$ exhibits the internal hom. Since $\mathrm{Fun}^L(\mathrm{Spc}, \mathcal{C}) \simeq \mathcal{C}$, it follows that the category of spaces Spc is the neutral element for the Lurie tensor product. Furthermore, one can show that $\mathcal{C} \otimes \mathcal{D} \simeq \mathrm{Fun}^R(\mathcal{C}^{\mathrm{op}}, \mathcal{D})$.

We will need a stable version of this observation.

Construction 1.9. Let Sp denote the category of spectra, defined as the colimit

$$\mathrm{Sp} = \mathrm{colim}(\mathrm{Spc}_* \xrightarrow{\Sigma} \mathrm{Spc}_* \xrightarrow{\Sigma} \dots)$$

in Pr^L , or dually, as the limit

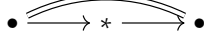
$$\mathrm{Sp} \simeq \lim(\dots \xrightarrow{\Omega} \mathrm{Spc}_* \xrightarrow{\Omega} \mathrm{Spc}_*)$$

in Pr^R , the category of presentable categories and right adjoints. Here Spc_* denotes the category of pointed spaces, whereas Σ and Ω denote the suspension and loop functor, respectively. By adjoining a point to an space and then “infinitely suspending it” we obtain a colimit preserving functor $\Sigma_+^{\infty} : \mathrm{Spc} \rightarrow \mathrm{Sp}$.

Remark 1.10. Let $\Sigma_+^{\infty} : \mathrm{Spc} \rightarrow \mathrm{Sp}$ be the suspension spectrum functor. Notice that a presentable category $\mathcal{C}$ is stable if and only if the canonical map $\mathrm{id}_{\mathcal{C}} \otimes \Sigma_+^{\infty} : \mathcal{C} \rightarrow \mathcal{C} \otimes \mathrm{Sp}$ is an equivalence. In particular, since Sp is a stable category, the inverse of the equivalence $\mathrm{Sp} \rightarrow \mathrm{Sp} \otimes \mathrm{Sp}$ makes Sp into a commutative algebra of Pr^L , whose unit is given by the sphere spectrum $\mathbb{S}$. It follows that the category of stable presentable categories and left adjoints can be realized as $\mathrm{Pr}_{\mathrm{st}}^L \simeq \mathrm{Mod}_{\mathrm{Sp}}(\mathrm{Pr}^L)$ and that taking spectrum objects $- \otimes \mathrm{Sp} : \mathrm{Pr}^L \rightarrow \mathrm{Pr}_{\mathrm{st}}^L$ defines a symmetric monoidal functor.

2. RETRACTS AND IDEMPOTENT COMPLETION

Notation 2.1. We let Idem^+ denote the following category



We also let Idem to be the full subcategory of Idem^+ spanned by the object lying in the middle (so that Idem consists of an object $*$ together with a map $e : * \rightarrow *$ such that $e \circ e \simeq e$).

Given a category $\mathcal{C} \in \text{Cat}$ we will refer to a functor $\text{Idem} \rightarrow \mathcal{C}$ as an *idempotent*. An idempotent $\text{Idem} \rightarrow \mathcal{C}$ is called *effective* if it extends to a map $\text{Idem}^+ \rightarrow \mathcal{C}$.

Definition 2.2. A category is called *idempotent-complete* if every idempotent is effective.

Remark 2.3. A retract closed full subcategory of an idempotent-complete category is idempotent-complete. Indeed, if $\mathcal{C} \subseteq \mathcal{D}$ is a retract closed full subcategory of an idempotent one, and $e : x \rightarrow x$ is an idempotent in $\mathcal{C}$, then the idempotent e is effective in $\mathcal{D}$, being the latter idempotent-complete.

Thus there exist an object $y \in \mathcal{D}$ and maps $y \xrightarrow{i} x \xrightarrow{p} y$ in $\mathcal{D}$ such that $pi \simeq \text{id}_y$ and $ip \simeq e$. In particular, y is a retract of x in $\mathcal{D}$. Since $x \in \mathcal{C}$ and $\mathcal{C}$ is closed under retracts, we have $y \in \mathcal{C}$. Since $\mathcal{C} \subseteq \mathcal{D}$ is full, the maps $i : y \rightarrow x$ and $p : x \rightarrow y$ belong to $\mathcal{C}$. Therefore the above retraction data lie in $\mathcal{C}$, and hence the idempotent e is effective in $\mathcal{C}$.

Remark 2.4. Every category admitting telescopes is idempotent-complete. Indeed, let $e : x \rightarrow x$ be an idempotent. Consider the sequential diagram $x \xrightarrow{e} x \xrightarrow{e} x \xrightarrow{e} \dots$ and let

$$y := \text{colim}(x \xrightarrow{e} x \xrightarrow{e} x \rightarrow \dots).$$

The colimit maps induce a morphism $p : x \rightarrow y$. On the other hand, the maps $e : x \rightarrow x$ at every stage are compatible with the transition maps, because $e^2 \simeq e$, and therefore induce a morphism $i : y \rightarrow x$. Then one has

$$ip \simeq e, \quad pi \simeq \text{id}_y.$$

Thus e is split. Hence every idempotent is effective.

We now show how to turn a category into an idempotent one.

Construction 2.5. Let $\mathcal{C} \in \text{Cat}$ be a category. Let $\text{PSh}(\mathcal{C}) = \text{Fun}(\mathcal{C}^{op}, \text{Spc})$ be the presheaf category and let $\mathfrak{y}_{\mathcal{C}} : \mathcal{C} \rightarrow \text{PSh}(\mathcal{C})$ be the Yoneda embedding. Define $\mathcal{C}^{\natural} \subseteq \text{PSh}(\mathcal{C})$ to be the full subcategory spanned by retracts of representables, that is by objects P for which there exists $X \in \mathcal{C}$ and maps $P \rightarrow \mathfrak{y}_{\mathcal{C}}(X) \rightarrow P$ whose composite is equivalent to id_P . We call $\mathcal{C}^{\natural}$ the *idempotent completion* of $\mathcal{C}$.

Proposition 2.6. Let $\mathcal{C} \in \text{Cat}$ be a category. Then inclusion $\mathfrak{y}_{\mathcal{C}} : \mathcal{C} \rightarrow \mathcal{C}^{\natural}$ is fully faithful and the category $\mathcal{C}^{\natural}$ is idempotent complete. Furthermore, precomposition with the Yoneda embedding induces an equivalence

$$(\mathfrak{y}_{\mathcal{C}})^* : \text{Fun}(\mathcal{C}^{\natural}, \mathcal{D}) \rightarrow \text{Fun}(\mathcal{C}, \mathcal{D})$$

for every idempotent complete $\mathcal{D}$.

Proof. The Yoneda embedding is fully faithful, hence so is its restriction to $\mathcal{C}^{\natural}$. The presheaf category $\text{PSh}(\mathcal{C})$ admits all small colimits, hence is idempotent complete by Remark 2.4. Since $\mathcal{C}^{\natural}$ is, by definition, closed under retracts inside $\text{PSh}(\mathcal{C})$, it is itself idempotent complete by Remark 2.3.

For the universal property, let $\mathcal{D}$ be idempotent complete and let $F : \mathcal{C} \rightarrow \mathcal{D}$ be any functor. By left Kan extension along $\mathfrak{y}_{\mathcal{C}}$ we obtain $\overline{F} : \text{PSh}(\mathcal{C}) \rightarrow \mathcal{D}$, and since $\mathcal{D}$ is idempotent complete the functor $\overline{F}$ carries retracts of representables to retracts of the corresponding values of F . Thus $\overline{F}$ factors essentially uniquely through $\mathcal{C}^{\natural}$, yielding an extension $\mathcal{C}^{\natural} \rightarrow \mathcal{D}$. Conversely, any functor $\mathcal{C}^{\natural} \rightarrow \mathcal{D}$ is determined by restriction to $\mathcal{C}$ because every object of $\mathcal{C}^{\natural}$ is a retract of some $\mathfrak{y}_{\mathcal{C}}(X)$. $\square$

Remark 2.7. Let $\mathcal{C} \in \text{Cat}^{\text{add}}$ be an additive category. The functor $\mathfrak{y}_{\mathcal{C}} : \mathcal{C} \rightarrow \mathcal{C}^{\natural}$ exhibits $\mathcal{C}$ as a *dense* subcategory of $\mathcal{C}^{\natural}$, that is, every object of $\mathcal{C}^{\natural}$ is a direct summand of an object of $\mathcal{C}$.

3. 2-RINGS

We now study idempotent-completeness related to stable categories.

Definition 3.1. We let $\text{Cat}^{\text{perf}} \subseteq \text{Cat}^{\text{st}}$ denote the full subcategory spanned by the idempotent-complete stable categories.

Definition 3.2. Let $f : \mathcal{C} \rightarrow \mathcal{D}$ in Cat^{st} be an exact functor. We will say that f is a *Karoubi equivalence* if the induced functor $f^{\natural} : \mathcal{C}^{\natural} \rightarrow \mathcal{D}^{\natural}$ is an equivalence.

In other terms, f is a Karoubi equivalence if it is fully faithful and its essential image is dense, in the sense of [Remark 2.7](#), that is every object of $\mathcal{D}$ is a retract of some object in the essential image of f .

Proposition 3.3. There exists a Bousfield localization $(-)^{\natural} : \text{Cat}^{\text{st}} \rightleftarrows \text{Cat}^{\text{perf}} : \text{incl}$ at the class of Karoubi equivalences.

Proof. Let $\mathcal{C} \in \text{Cat}^{\text{st}}$. Its idempotent completion $\mathcal{C}^{\natural}$ is again stable: indeed, it is the full subcategory of $\text{PSh}(\mathcal{C})$ spanned by retracts of representables, and this class is closed under finite limits and finite colimits. Let $\mathcal{D} \in \text{Cat}^{\text{perf}}$. By the universal property of idempotent completion, restriction along the Yoneda embedding induces an equivalence

$$\mathfrak{z}^* : \text{Fun}(\mathcal{C}^{\natural}, \mathcal{D}) \rightarrow \text{Fun}(\mathcal{C}, \mathcal{D}).$$

Since the Yoneda embedding $\mathcal{C} \rightarrow \mathcal{C}^{\natural}$ is exact and $\mathcal{D}$ is stable, this equivalence restricts to an equivalence

$$\text{Fun}^{\text{ex}}(\mathcal{C}^{\natural}, \mathcal{D}) \simeq \text{Fun}^{\text{ex}}(\mathcal{C}, \mathcal{D}).$$

Thus $(-)^{\natural}$ is left adjoint to the inclusion $\text{Cat}^{\text{perf}} \hookrightarrow \text{Cat}^{\text{st}}$. Finally, a functor $f : \mathcal{C} \rightarrow \mathcal{D}$ is inverted by this localization if and only if $f^{\natural} : \mathcal{C}^{\natural} \rightarrow \mathcal{D}^{\natural}$ is an equivalence. By definition, these are precisely the Karoubi equivalences. $\square$

We record a convenient strengthening of the above (proved in [\[CDH⁺25, Proposition A.3.4\]](#)).

Remark 3.4. Let $\mathcal{C} \in \text{Cat}^{\text{st}}$ be a stable category and let $K_0(\mathcal{C})$ be its Grothendieck group. Define the *minimalisation* $\mathcal{C}_{\min} \subseteq \mathcal{C}$ to be the full stable subcategory spanned by objects X with $[X] = 0$ in $K_0(\mathcal{C})$. We call $\mathcal{C}$ *minimal* if $\mathcal{C}_{\min} = \mathcal{C}$.

Then localising Cat^{st} at the class of Karoubi equivalences yields both a left and a right Bousfield localisation. The left-local objects are the minimal stable categories, and the right-local objects are the idempotent complete stable categories. Moreover, the left and right adjoints of this localisation are given by $\mathcal{C} \mapsto \mathcal{C}_{\min}$ and $\mathcal{C} \mapsto \mathcal{C}^{\natural}$, respectively. In particular, an exact functor is a Karoubi equivalence if and only if it induces an equivalence on minimalisations, and if and only if it induces an equivalence on idempotent completions. Consequently, the functor $(-)^{\natural} : \text{Cat}^{\text{st}} \rightarrow \text{Cat}^{\text{perf}}$ preserves both limits and colimits.

We now show that Cat^{perf} may be recovered inside $\text{Pr}_{\text{st}}^{\text{L}}$.

Proposition 3.5. There is an equivalence $\text{Ind} : \text{Cat}^{\text{perf}} \rightleftarrows \text{Pr}_{\text{st}}^{\text{L}, \omega} : (-)^{\omega}$.

Proof. It is clear that the ind-completion and taking compact objects sit into an adjunction $\text{Ind} : \text{Cat}^{\text{perf}} \rightleftarrows \text{Pr}_{\text{st}}^{\text{L}, \omega} : (-)^{\omega}$, thus it suffices to show that unit and counit are equivalences. Now:

- (1) Let $\mathcal{C} \in \text{Cat}^{\text{perf}}$ and consider the unit $\mathcal{C} \rightarrow (\text{Ind}(\mathcal{C}))^{\omega}$. It suffices to show that every compact object of $\text{Ind}(\mathcal{C})$ lies in the essential image of $\mathfrak{z}_{\mathcal{C}}$. Let $X \in \text{Ind}(\mathcal{C})$ be compact. By definition of $\text{Ind}(\mathcal{C})$, the object X can be written as a filtered colimit $X \simeq \text{colim}_{i \in I} \mathfrak{z}_{\mathcal{C}}(c_i)$ with $c_i \in \mathcal{C}$. Consider the identity map $\text{id}_X : X \rightarrow X \simeq \text{colim}_i \mathfrak{z}_{\mathcal{C}}(c_i)$. Since X is compact, id_X factors through some stage, so there exists $i \in I$ and a map $u : X \rightarrow \mathfrak{z}_{\mathcal{C}}(c_i)$ such that the composite

$X \xrightarrow{u} \mathfrak{L}_{\mathcal{C}}(c_i) \rightarrow \operatorname{colim}_i \mathfrak{L}_{\mathcal{C}}(c_i) \simeq X$ is equivalent to id_X . Thus X is a retract of $\mathfrak{L}_{\mathcal{C}}(c_i)$ in $\operatorname{Ind}(\mathcal{C})$.

Since $\mathfrak{L}_{\mathcal{C}}$ is fully faithful, a retract data on $\mathfrak{L}_{\mathcal{C}}(c_i)$ correspond to retract data on c_i in $\mathcal{C}$. Since $\mathcal{C}$ is idempotent-complete, this retract is represented by an object $c \in \mathcal{C}$ and an equivalence $X \simeq \mathfrak{L}_{\mathcal{C}}(c)$. Therefore every compact object of $\operatorname{Ind}(\mathcal{C})$ lies in $\mathfrak{L}_{\mathcal{C}}(\mathcal{C})$, and we conclude that the induced functor $\mathcal{C} \rightarrow (\operatorname{Ind}(\mathcal{C}))^\omega$ is essentially surjective. It is fully faithful since $\mathfrak{L}_{\mathcal{C}}$ is fully faithful, hence it is an equivalence.

- (2) For the counit, let $\mathcal{D} \in \operatorname{Pr}_{\operatorname{st}}^{L,\omega}$ and consider $\operatorname{Ind}(\mathcal{D}^\omega) \rightarrow \mathcal{D}$. The inclusion $i : \mathcal{D}^\omega \hookrightarrow \mathcal{D}$ extends (by the universal property of ind-completion) to a colimit-preserving functor $\varepsilon : \operatorname{Ind}(\mathcal{D}^\omega) \rightarrow \mathcal{D}$ whose restriction to $\mathcal{D}^\omega$ is i . It suffices to show that ε is an equivalence. First, ε is essentially surjective since $\mathcal{D}$ is compactly generated by $\mathcal{D}^\omega$, and ε preserves colimits and contains $\mathcal{D}^\omega$ in its image. For full faithfulness, it suffices to check it on compact generators: given $x, y \in \mathcal{D}^\omega$ the equivalence $\operatorname{Hom}_{\operatorname{Ind}(\mathcal{D}^\omega)}(\mathfrak{L}(x), \mathfrak{L}(y)) \simeq \operatorname{Hom}_{\mathcal{D}^\omega}(x, y) \simeq \operatorname{Hom}_{\mathcal{D}}(x, y)$ shows that ε is fully faithful on the subcategory of compact objects. Since both source and target are generated under colimits by these compact objects and ε preserves colimits, it follows that ε is fully faithful on all objects. Hence ε is an equivalence.

Hence the claim. $\square$

Using this result, we can equip $\operatorname{Cat}^{\operatorname{perf}}$ with a monoidal structure.

Remark 3.6. Proceed as in Remark 1.8: the category $\operatorname{Cat}^{\operatorname{st}}$ admits a symmetric monoidal structure characterized as follows. If $\mathcal{C}, \mathcal{D} \in \operatorname{Cat}^{\operatorname{st}}$, then $\mathcal{C} \otimes_{\operatorname{Cat}^{\operatorname{st}}} \mathcal{D}$ is the stable category equipped with an exact functor $\mathcal{C} \times \mathcal{D} \rightarrow \mathcal{C} \otimes_{\operatorname{Cat}^{\operatorname{st}}} \mathcal{D}$ which is exact separately in each variable and such that, for every stable category $\mathcal{E}$, precomposition induces an equivalence

$$\operatorname{Fun}^{\operatorname{ex}}(\mathcal{C} \otimes_{\operatorname{Cat}^{\operatorname{st}}} \mathcal{D}, \mathcal{E}) \rightarrow \operatorname{Fun}^{\operatorname{ex}, \operatorname{ex}}(\mathcal{C} \times \mathcal{D}, \mathcal{E}).$$

Here $\operatorname{Fun}^{\operatorname{ex}, \operatorname{ex}}(\mathcal{C} \times \mathcal{D}, \mathcal{E})$ denotes the full subcategory of $\operatorname{Fun}(\mathcal{C} \times \mathcal{D}, \mathcal{E})$ spanned by those functors which are exact separately in each variable. Using idempotent completion, this gives a symmetric monoidal structure on $\operatorname{Cat}^{\operatorname{perf}}$ by setting

$$\mathcal{C} \otimes_{\operatorname{Cat}^{\operatorname{perf}}} \mathcal{D} := (\mathcal{C} \otimes_{\operatorname{Cat}^{\operatorname{st}}} \mathcal{D})^{\mathfrak{h}}.$$

Then:

Proposition 3.7. The equivalence $\operatorname{Ind} : \operatorname{Cat}^{\operatorname{perf}} \simeq \operatorname{Pr}_{\operatorname{st}}^{L,\omega} : (-)^\omega$ is symmetric monoidal.

Proof. Let $\mathcal{E} \in \operatorname{Pr}_{\operatorname{st}}^L$. Then

$$\begin{aligned} \operatorname{Fun}^L(\operatorname{Ind}(\mathcal{C} \otimes_{\operatorname{Cat}^{\operatorname{perf}}} \mathcal{D}), \mathcal{E}) &\simeq \operatorname{Fun}^{\operatorname{ex}}(\mathcal{C} \otimes_{\operatorname{Cat}^{\operatorname{perf}}} \mathcal{D}, \mathcal{E}) \\ &\simeq \operatorname{Fun}^{\operatorname{ex}}(\mathcal{C} \otimes_{\operatorname{Cat}^{\operatorname{st}}} \mathcal{D}, \mathcal{E}) \\ &\simeq \operatorname{Fun}^{\operatorname{ex}, \operatorname{ex}}(\mathcal{C} \times \mathcal{D}, \mathcal{E}). \end{aligned}$$

The first equivalence is the universal property of Ind-completion, the second follows because $\mathcal{E}$ is idempotent-complete and the third is the universal property of $\mathcal{C} \otimes_{\operatorname{Cat}^{\operatorname{st}}} \mathcal{D}$. On the other hand,

$$\begin{aligned} \operatorname{Fun}^L(\operatorname{Ind}(\mathcal{C}) \otimes \operatorname{Ind}(\mathcal{D}), \mathcal{E}) &\simeq \operatorname{Fun}^{L,L}(\operatorname{Ind}(\mathcal{C}) \times \operatorname{Ind}(\mathcal{D}), \mathcal{E}) \\ &\simeq \operatorname{Fun}^{\operatorname{ex}, \operatorname{ex}}(\mathcal{C} \times \mathcal{D}, \mathcal{E}), \end{aligned}$$

where the first equivalence is the universal property of the Lurie tensor product, and the second follows by applying the universal property of Ind-completion separately in the two variables. Thus both $\operatorname{Ind}(\mathcal{C} \otimes_{\operatorname{Cat}^{\operatorname{perf}}} \mathcal{D})$ and $\operatorname{Ind}(\mathcal{C}) \otimes \operatorname{Ind}(\mathcal{D})$ corepresent the same functor of $\mathcal{E} \in \operatorname{Pr}_{\operatorname{st}}^L$. Hence they are equivalent by Yoneda. $\square$

Corollary 3.8. The Lurie tensor product of two ω -presentable stable presentable categories is again ω -presentable. Since the category of spectra is ω -presentable, it follows that $\mathrm{Pr}_{\mathrm{st}}^{\mathrm{L},\omega} \hookrightarrow \mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}}$ is symmetric monoidal.

We can finally take commutative algebra objects:

Definition 3.9. We denote by $2\mathrm{CAlg} = \mathrm{CAlg}(\mathrm{Cat}^{\mathrm{perf}})$ the category of 2-rings.

Explicitly, a 2-ring consist of a small idempotent complete stable category $\mathcal{A}$ together with a symmetric monoidal structure for which the tensor product $- \otimes - : \mathcal{A} \times \mathcal{A} \rightarrow \mathcal{A}$ is exact in each variable separately.

4. EXAMPLES

We conclude by discussing some examples.

Example 4.1. Let X be a quasi-compact quasi separated scheme. Then the category of perfect complexes $\mathrm{Perf}(X)$ defines a 2-ring.

Example 4.2. Let $R \in \mathrm{CAlg}(\mathrm{Sp})$ be a commutative ring spectrum. We denote by Perf_R the full subcategory of Mod_R spanned by the compact R -modules. Since Mod_R is a stable presentable symmetric monoidal category and the tensor product preserves compact objects, Perf_R is a small idempotent-complete stable symmetric monoidal category. Hence Perf_R defines a 2-ring.

In particular, for $R = \mathbb{S}$ one obtains

$$\mathrm{Perf}_{\mathbb{S}} = \mathrm{Sp}^{\omega},$$

the category of finite spectra. This is the initial 2-ring.

Example 4.3. Let $R \in \mathrm{CAlg}(\mathrm{Sp})$ be a commutative ring spectrum and let G be a finite group. Define

$$\mathrm{Rep}(G, R) := \mathrm{Fun}(BG, \mathrm{Perf}_R).$$

Then $\mathrm{Rep}(G, R)$ is a small idempotent-complete stable category. Moreover, the pointwise tensor product endows $\mathrm{Rep}(G, R)$ with a symmetric monoidal structure, which is exact in each variable. Hence $\mathrm{Rep}(G, R)$ defines a 2-ring.

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